

[Dashboard](#) / [Methodology](#)

Methodology

How Pharos grades stablecoins: transparent scoring across safety, peg stability, liquidity, yield, and contagion risk. Treat this page like a reference manual, not a marketing explainer. All scoring methodologies operate over the active subset of tracked stablecoins; pre-launch and frozen coins are excluded from new computations and live aggregates.

READER GUIDE

Reader mode keeps summaries up front. Switch to Analyst for formulas, caveats, and worked examples.

VIEW

Reader

Analyst

Page rhythm: summary, quick facts, worked example, technical notes.

Summary

Model purpose and the core signal to scan first.

Quick Facts

Cadence, score range, dependencies, and failure behavior.

Worked Examples

Real inputs run through the same functions used in production.

Technical Notes

Expanded formulas, caveats, and changelog links when you need them.

New to stablecoin design? [Learn how each stablecoin design produces its peg](#) before the scoring formulas.

EDITORIAL PREFACE

How Pharos grades every stablecoin, and what it refuses to grade.

Every safety grade Pharos publishes is the answer to one question: if this stablecoin started bleeding tomorrow, how much of the loss would the holder eat before the system stopped it? Four base dimensions answer different parts of that question. Liquidity / Exit (30%) measures whether you can leave at all. Dependency Risk (25%) measures whether something else has to hold for this one to hold. Resilience (20%)

measures the quality of what backs the coin and who is allowed to touch it. Decentralization (15%) measures whether anyone can switch the lights off. We weight them in that order because, in every real depeg we have studied, the exit was already gone before the press release went out.

The peg score does not sit alongside the four base dimensions. It multiplies them. After the weighted base score is computed, we apply $(\text{pegScore} / 100)^{0.40}$ as a power curve. The exponent is deliberate. At 1.0, a ten-point drop in peg score would erase ten points of safety, which punishes mid-quality pegs more than the data supports. At 0.0, the peg would not constrain the grade at all, which would let a structurally fine asset with a chronically wobbly peg coast on collateral quality. Forty splits the difference: a peg score of 90 takes about four percent off; a peg score of 10 takes sixty percent off. Strong pegs are nearly untouched, broken pegs are sharply demoted, and the curve does the work without a cliff.

Dependency Risk is the dimension that most often surprises issuers. A wrapper that lives entirely inside a parent stablecoin cannot, by construction, be safer than its parent. We enforce that with a variant overall cap, not by averaging the parent in. Averaging would let a well-built wrapper around a weak base settle at some middle grade that no one should rely on; capping says the downstream cannot escape the upstream, which is how the dependency actually behaves under stress. The same logic produces family-specific wrapper ceilings for risk-absorption, strategy vaults, and bond-maturity variants. The cap is honest about what the asset is: a claim on something else.

Yield tokens get flagged, not folded. Pharos tracks yield-bearing and NAV tokens through a separate yield-risk pipeline because the failure modes are different in kind, not just degree. A holder of a \$1.00

stablecoin and a holder of a \$1.04 wrapper are not running the same trade. The wrapper carries source risk, sustainability risk, and benchmark drift that have no analog in the underlying peg. Until those signals are sourced to a standard we will publish against, we surface them on the yield page and leave the safety score honest about what it does and does not measure. Configured NAV wrappers can still inherit peg risk from a referenced base; pure fund-share tokens with no peg reference receive a peg multiplier of 1.0 and are graded on the structural dimensions alone.

— Pharos, May 2026

Jump to Section

Lifecycle Phases

Pricing Pipeline

Stability Index

Safety Scores

Infrastructure

Liquidity Score

Mint/

Stablecoin Lifecycle Phases

Lifecycle phase is a data-collection policy. No per-domain methodology version is bumped when a coin transitions between phases.

Every tracked stablecoin is in one of three lifecycle phases. The phase determines which surfaces ingest, score, and read back the coin's data. Scoring methodologies always operate over the active subset; pre-launch and frozen coins are excluded from new computations and live aggregates.

Active

Default state. Full worker ingestion (supply, price, blacklist, mint/burn, DEX liquidity, yield), full inclusion in PSI, DEWS, Safety Scores, Liquidity Score, Bank Run Gauge, and all live aggregates. Listed in the homepage table, search, compare picker, and sitemap.

Pre-launch

Tracked for visibility before launch. No live data yet, so the coin is excluded from score computations and live aggregates. Surfaced on [/upcoming](#) with milestone metadata and on its own pre-launch detail page; not in the homepage table, active taxonomy pages, portfolio picker, or live comparison picker.

Frozen

The coin is effectively defunct (issuer abandonment, supply trending to zero, irrecoverable depeg, regulatory shutdown) but its historical data and detail page are preserved as an archive. No new data is collected, and the coin is excluded from PSI, DEWS, Safety Score recomputations, Bank Run Gauge, and other live aggregates. Frozen entries appear in the [cemetery](#) with an "archived data available" link to the preserved detail page. The report card shows an all-F stub matching the existing dead-stablecoin pattern.

TRACKED

Every tracked coin (active + pre-launch + frozen). Used for canonical-ID lookups, registry validation, and static stablecoin detail routes.

ACTIVE

Active subset only. Used by every write-side cron and live aggregate.

READABLE

Active + frozen. Used by readback/archive surfaces that should preserve frozen assets while excluding pre-launch assets.

FROZEN

Frozen subset only. Drives the cemetery merge and the dataset export.

See also: [Cemetery](#) · [Upcoming launches](#) · Operator runbook: [docs/freezing-stablecoins.md](#)

Pricing Pipeline Methodology v6.06 [Version history →](#)

Version increments when price sources, consensus algorithm, enrichment passes, or validation rules change.

Every score Pharos computes starts with a price. The pricing pipeline collects quotes from more than a dozen live voices, requires fully pairwise agreement inside each cluster, and publishes the highest-confidence result with explicit source-freshness semantics.

Source diversity. Kraken and Bitstamp extend the direct venue set. Fresh RedStone prices need timestamped multi-venue breakdowns. The primary promoted DEX bridge now spans Fluid, Balancer, Curve, Raydium, Orca, Meteora, PancakeSwap, Aerodrome Slipstream, and Velodrome Slipstream, with structured source-filter telemetry and DEX lane confidence profiles attached when those lanes participate. Targeted exact-address augmentation can add DexScreener, DexPaprika, GeckoTerminal, Alchemy Prices, Moralis, and Solana Birdeye quotes for assets with missing prices or below-target source depth. DEX bridge and address-provider identity are canonical-only at runtime, so addressed unknown tokens are dropped instead of being reinterpreted by symbol, promoted protocol DEX prices only enter consensus when they are corroborated or no non-DEX voices exist, and direct-API quote legs prefer tracked live stablecoin prices instead of unconditional \$1 symbol assumptions.

Pool challenge. A pool challenge guard downgrades confidence and replaces the price with a protocol-aware TVL-weighted median only when large DEX pools from at least two independent protocols diverge from soft consensus, with divergence evaluated from one TVL-weighted median per protocol so a single rogue pool cannot make an otherwise agreeing protocol count as corroborating disagreement, including DEX-inclusive soft clusters unless an exempt hard source is present. NAV tokens are excluded because their wide clustering threshold makes pool-level divergence a poor signal for redeemable-at-NAV assets. When the guard replaces a price, the replacement is now also reflected in the per-source candidate list so downstream corroboration continuity uses the replacement mark rather than the superseded candidate. Corroborated severe downside from multiple candidate sources including a depeg-authoritative source can be downgraded by pool challenge, but its price is preserved and the same evidence satisfies the temporal-jump guard. Dead blocked DEX slugs such as Bunni are excluded upstream and never qualify as challenger or DEX-bridge inputs.

Overrides & FX. Protocol-level redemption prices override market data for wrapper assets. Chainlink refreshes supported FX and commodity reference rates, and the dated secondary FX mirror can temporarily carry the wider fiat reference stack when Frankfurter is unavailable, with ExchangeRate-API as a tertiary daily fallback if both primary FX paths are down. If those live FX fetches still fail but the last published daily references remain within cadence, Pharos carries those dated references forward as a healthy refresh. Even cached-fallback FX runs keep probing the independent OXR, Chainlink, and metals paths, so a recovered intraday subset can promote the lane back to live without waiting for the full Frankfurter stack.

Freshness tracking. The live payload distinguishes true upstream observation timestamps from locally stamped fetch-time freshness via `priceObservedAtMode`, so a hard single-source print only becomes depeg-authoritative when its freshness is source-native rather than inferred from local collection time. The source registry now also records each provider's freshness kind, maximum trusted age, and whether it truly supports upstream timestamps. CoinGecko simple-price rows use upstream `last_updated_at` when it is present and are rejected when that upstream timestamp is outside the trusted freshness window. Bitstamp, Coinbase, and Curve on-chain reads now publish upstream-observed freshness provenance rather than stamping rows at local fetch time, and the `crvUSD` Curve oracle feed enforces a 5-minute on-chain staleness guard using the aggregator block timestamp and records against its own dedicated circuit breaker. Replay cache now enforces each source's native max trusted age alongside the composite 6-hour cap.

Native-peg corroboration. For supported non-USD fiat assets with a reliable native-market quote, the pipeline now derives a fresh `native quote × FX reference USD` mark during post-enrichment. That lane can correct materially divergent weak or mixed-source live USD publications, fill a missing live price for supported assets, and still veto or resolve downstream depeg mutations when the direct native pair disagrees with a derived `USD price / FX reference move`. It remains a fresh validation lane rather than a replay-safe primary consensus voice, so it does not become cached continuity on later runs.

Historical replay parity. Supported non-USD fiat backfills now prefer direct CoinGecko native-fiat history and compare that series to the native 1.0 peg before falling back to USD-denominated market history. That removes long synthetic depeg streaks created only by replay-time USD / FX mismatch.

Enrichment & confidence. A 6-pass enrichment pipeline fills gaps for long-tail coins. Each asset is tagged with a confidence level so downstream systems can react to data quality, and severe fixed-peg downside publication now requires corroboration unless it comes from an explicit protocol redemption or pool-challenge replacement mark. Two-source clusters composed only of list-style aggregators (CoinGecko, DefiLlama, DefiLlama-list) are now downgraded to single-source regardless of which two combine, closing the CoinGecko + DefiLlama-detail tautology alongside the existing CG + DL-list downgrade. Cluster tiebreak also prefers hard-tier clusters over equal-weight soft-tier clusters before falling through to spread and peg-proximity rules. A lone promoted DEX protocol is admitted only when no non-DEX source exists, or when a hard market/oracle/protocol source agrees within threshold. Corroborating candidate prices now stay attached through later validation when the selected primary result is unchanged, so a real severe depeg is not cleared as a single-source mark. When a confirmed severe depeg briefly loses corroboration, the pipeline preserves trusted continuity from fresh replay-safe `price_cache` rows instead of letting the asset flap to `N/A`. DefiLlama contract fallbacks must now pass the same peg-aware plausibility gates before they can resolve an asset, and DexScreener symbol search is retired so addressless assets are not probed through noisy symbol-only identity. DefiLlama `/coins` contract-price fallback and DexScreener dex-liquidity and dex-discovery paths now record against their own dedicated circuit breakers instead of reusing unrelated breaker state, and the same provider diagnostics and GeckoTerminal probe statistics collected during each run are now surfaced on `/api/status` for operator visibility.

UPDATE CADENCE

15m refresh

SOURCES

20+ live voices

OUTPUT

Price + confidence tag per asset

Preconditions & Failure Modes

MINIMUM DATA

At least 1 source must return a price; consensus requires 2+ for high confidence

CIRCUIT BREAKERS

Most live upstream families are breaker-gated: opens after 3 failures, probes every 30 min

FAILURE BEHAVIOR

Degraded sources are excluded from consensus; enrichment pipeline fills remaining gaps; stale cache used as last resort

► Worked example: USDC price consensus across 7 sources

► Technical details: source weights, consensus algorithm, overrides, enrichment, and validation

Stability Index Methodology v3.3 [Version history →](#)

Version increments when PSI formula, caps, bands, component definitions, or other score-affecting input semantics change.

The Pharos Stability Index (PSI) is a market-level 0–100 health score for the stablecoin ecosystem. It is recomputed every 30 minutes from live depeg conditions and stress signals, then aggregated into daily history snapshots.

See also: [PegScore + DEWS](#) · [Safety Scores](#)

UPDATE CADENCE

30m refresh

SCORE RANGE

0-100 market health

MAIN USE

Bands: BEDROCK to
MELTDOWN

Preconditions & Failure Modes

MINIMUM DATA

Scorer accepts empty depeg and empty DEWS-stress row sets, but the cron requires total market cap > 0 plus readable active-depeg and DEWS input queries

REQUIRED SOURCES

Market-cap totals, active depeg inputs (current stablecoins price, or recent replay-safe price_cache fallback for already-open depegs), and a readable DEWS stress-signals table

FAILURE BEHAVIOR

Returns null when market-cap input is missing/ ≤ 0 ; the cron also skips publication when active-depeg or DEWS inputs are unavailable, and the API serves the last valid value

HISTORICAL REPLAY

Backfills score any depeg overlapping the UTC day, canonicalize legacy depeg IDs into the current PSI universe, use same-day supply_history prices when they capture the move, cap replayed daily deviation at the event peak, and keep peak event deviation as a start-day floor only when the depeg stayed open through the UTC close and the daily snapshot misses the move

► **Worked example (verified against computeStabilityIndex)**

► **Technical details: formula, component math, depeg handling, and condition bands**

Safety Scores Grading Methodology v7.26 [Version history →](#)

Version increments when weights, thresholds, or dimension definitions change.

Pharos synthesizes multiple data signals into a single transparent grade per stablecoin. The overall score is computed in two steps: first, a weighted average of four base dimensions (exit liquidity, resilience, decentralization, dependency risk), then a peg stability multiplier that penalizes coins with poor pegs while barely affecting well-pegged ones. The exit-liquidity dimension blends raw DEX liquidity with redemption-backstop quality only when the route has usable current evidence. Reserve data is a separate resilience input: live reserve sync can improve collateral quality only when the latest snapshot is fresh, independent, clean, and score-grade. When some base dimensions lack data (NR), their weight is redistributed proportionally among rated ones. Active depeg caps use the open event's peak deviation, while the peg dimension itself remains a direct pegScore passthrough.

See also: [PegScore + DEWS](#) · [Liquidity Score](#) · [Infrastructure](#)

MODEL SHAPE

4 dimensions + peg multiplier

GRADE OUTPUT

A+ to F, with NR

KEY CAVEAT

No exit signal = 10% penalty

Preconditions & Failure Modes

MINIMUM DATA

At least 2 rated non-peg dimensions

REQUIRED SOURCES

Peg summary, DEX liquidity/redemption data, and dependency/metadata inputs

FAILURE BEHAVIOR

NR if peg is missing on non-NAV coins; no-liquidity penalty when both DEX and redemption evidence are unavailable; live reserve adapters stay detail-visible when they are stale, degraded, or proof-only

Three Reserve-Related Signals

These labels are easy to conflate. They do different jobs, and only two can affect the Safety Score.

Signal	What it means	Score use
Reserve view	A detail-page reserve display exists: live, curated-validated, proof, curated, or estimated.	Informational unless it also passes the score-grade live-reserve gates.
Score-grade live reserve	The current report-card snapshot used a fresh, clean, independent live reserve snapshot for collateral quality.	Can replace curated collateral slices inside Resilience.
Redemption telemetry	A live reserve adapter emitted current redemption capacity, fee, freshness, or route-status metadata.	Can feed Redemption Backstop capacity or fee scoring; it does not automatically change collateral quality.

► **Worked example (verified against computeOverallGrade)**

► **Interactive calculator: explore how weights and thresholds shape the grade**

► **Technical details: full pipeline, dimension formulas, thresholds, and caveats**

Infrastructure Tagging v1.0

Infrastructure identifies the shared technical foundation a stablecoin was built on. Pharos currently recognises three values: Liquity v1, Liquity v2, and M0. The tag answers the question: *what shared technology does this coin inherit risk from?*

Liquity v1 and Liquity v2 are *code lineages*— coins that fork the original Liquity CDP implementation (v1) or its newer BOLD-style design (v2). Forks share source code but operate independently with their own reserves, governance, and Stability Pools. A vulnerability in the upstream Liquity codebase potentially affects every fork in that branch, even though the forks have no operational relationship.

M0 is an *issuance-platform lineage*— coins built on M0's smart-contract rails (minter governance, the SwapFacility, and the `MExtension.sol` contract pattern). M0 provides the issuance machinery; the reserve composition is set by the issuer and may or may not include the underlying \$M token. Some M0-built coins are simple \$M wrappers; others manage diversified collateral via M0's infrastructure. A governance issue at the M0 protocol level potentially affects every M0-built coin, even though their day-to-day operations and reserves are independent.

STORAGE

Array field on each
StablecoinMeta entry

CARDINALITY

Zero, one, or many
infrastructures per coin

SURFACES

Detail badge, homepage
filter, taxonomy pages,
methodology

Liquidity Score v5.7 [Version history →](#)

Version increments when liquidity formula weights, source inclusion rules, or TVL normalization logic changes.

Composite 0–100 score measuring DEX liquidity depth per stablecoin, updated every 30 minutes.
Aggregates pool data across all major DEXes and chains.

Dead or explicitly blocked DEX slugs such as Bunni are excluded upstream from crawl intake, retained pools, challenger snapshots, and DEX-implied price publication instead of being treated as low-quality live venues.

► Pool Matching & Deduplication

UPDATE CADENCE

30m refresh

SIGNAL MIX

5 weighted liquidity
components

OUTPUT

0-100 DEX depth score

Preconditions & Failure Modes

MINIMUM DATA

No hard minimum in scorer;
missing stability history
defaults to neutral 50 sub-
scores

REQUIRED SOURCES

Pool TVL/volume/chain data
plus mechanism and pair-
quality metadata

FAILURE BEHAVIOR

If both DEX liquidity and
eligible redemption evidence
are unavailable, the report-
card Liquidity / Exit
dimension is NR

► Worked example (verified against computeLiquidityScore)

► Technical details: component weights, TVL scaling, and quality adjustments

Mint/Burn Flow Scoring v6.11 [Version history →](#)

Version increments when flow scoring logic, tracked event semantics, or ingestion attribution policies change.

Pharos tracks on-chain mint and burn events for major stablecoins via Alchemy JSON-RPC (Transfer mints/burns plus USDT Issue/Redeem). These raw events are aggregated into hourly buckets and exposed as two separate signals: raw net flow for current direction, and a baseline-relative pressure score for context. Counted flow excludes bridge transfers, review-required burns, and atomic roundtrips.

DATA SOURCE

On-chain mint + burn events

PRIMARY SCORE

Pressure Shift vs 30D

MAIN OUTPUTS

Net flow, gauge, and FtQ

Preconditions & Failure Modes

MINIMUM DATA

Pressure Shift vs 30D
requires at least 7 days of
flow history per coin

REQUIRED SOURCES

24h mint/burn totals plus
30-day baseline aggregates

FAILURE BEHAVIOR

Pressure shift can be null
(NR); gauge is null when no
weighted inputs contribute;
FtQ needs $\pm \$100M$ dual
threshold

COUNTED ROWS

Economic-flow aggregates
count standard rows only,
which in practice means
non-bridge mints plus
effective burns

► **Worked example (verified against computeFlowIntensity)**

► **Technical details: two-signal pipeline, pressure formula, and gauge bands**

Yield Intelligence v8.17 [Version history →](#)

Version increments when APY source resolution, source arbitration, history semantics, PYS scoring logic, or eligibility rules for discovered yield sources change.

Pharos tracks native stablecoin yield plus selected lending opportunities and computes a risk-adjusted ranking via the Pharos Yield Score (PYS). Core rankings publish hourly using a source-aware APY resolution strategy, while slower supplemental-source families refresh on a separate four-hour lane. Alternative sources are retained when multiple valid yield paths exist, address-first identity is used before symbol fallback, curated exact-pool overrides can cover named non-stablecoin venues, confidence-weighted arbitration selects the primary row, and published lending suggestions exclude Resolv / USR-linked venues so broken wrapper ecosystems do not surface as recommended base-asset routes. Eligible tracked wrapper variants can also surface their native/wrapper sources as linked parent routes, so parent assets such as BOLD can show yBOLD and sBOLD context without removing the variants' own rows. The curated auto-discovery lane also pins current Felix, Sovryn, Loopscale, Resupply, Sovryn XUSD, and Anzens exact venues when they pass the normal source-quality gates, and the allowlist now includes reviewed Tier C venues such as AutoFinance, Neverland, Metrom, Mystic Finance, Bitway, and Frankencoin. Rate-derived coverage includes cgUSD, USDN, BENJI, WTGXX, USTBL, and EUTBL, while commodity exact-pool coverage includes XAUT on Lista Lending and PAXG on Hydration. Published lending-opportunity rows now also require observable venue TVL and a size floor of at least 0.1% of the tracked stablecoin's current supply before they can become the live recommendation. PYS is benchmark-aware and source-risk-aware, with missing source-risk evidence treated as neutral. Curve Savings crvUSD now follows the active on-chain profit-unlock stream instead of a trailing exchange-rate delta; pre-launch assets remain manifest-visible as intentional gaps but cannot publish into the live leaderboard before launch.

See also: [Safety Scores](#) · [Liquidity Score](#)

UPDATE CADENCE

1h publish / 4h supplemental

APY PRIORITY

Confidence-weighted across deterministic, curated, and fallback sources

OUTPUT

PYS (0-100)

Preconditions & Failure Modes

MINIMUM DATA

Need one resolved APY source; deterministic exchange-rate rows can publish from current state, but need prior source-specific history for a nonzero exchange-rate-derived APY

REQUIRED SOURCES

Direct on-chain reads, curated DeFiLlama pools, curated protocol-native APIs, rate-derived benchmark inputs, or 7-45d price history

FAILURE BEHAVIOR

No resolved source skips coin update; PYS returns 0 when $\text{apy30d} \leq 0$ or the benchmark-adjusted effective yield is non-positive (safety defaults to 40 / NR if live report-card hydration is missing; missing source-risk penalty resolves to neutral 1), while degraded benchmark or safety inputs are surfaced in provenance

Adapter manifest

The adapter manifest is the canonical machine-readable source list for every yield-bearing asset Pharos tracks. It enumerates the deterministic on-chain readers, curated DeFiLlama pools, protocol-API venues, rate-derived benchmark fallbacks, price-derived NAV fallbacks, auto-discovery lending overrides, and intentional coverage gaps that feed the ranking pipeline, along with each entry's lifecycle (`active`, `quarantined`, `intentional-gap`, `experimental`). The manifest is published at </api/yield-adapter-manifest> with a 5-minute cache, stamped with the current methodology version on every entry.

► **Worked example (verified against computePYS)**

► **Technical details: APY source resolution, confidence arbitration, PYS formula, NAV handling, and limits**

PegScore and Depeg Early Warning Score (DEWS) v6.0 [Version history →](#)

Version increments when depeg thresholds, confirmation policy, peg-score formula terms, or DEWS signal composition or score-affecting input semantics change.

PegScore observes the past and present by scoring realized peg behavior, while DEWS is forward-looking and scores pre-price and live-market depeg stress signals.

Depeg Tracker combines live event detection, secondary-source confirmation rules for large-cap assets, low-confidence primary prices, and extreme moves, plus a per-coin peg score that penalizes time off peg, event severity, active depegs, and unstable event spread. Pending depeg confirmation checks same-direction off-chain sources (CoinGecko or DefiLlama), supported native-peg quotes, Binance tickers, trusted aggregate DEX prices, and large challenger pools before promoting or rejecting candidates.

► **Depeg Confirmation & Trust Gates**

See also: [Mint/Burn Flow](#) · [Liquidity Score](#)

PEGSORE FOCUS

History: realized peg behavior

DEWS FOCUS

Forward stress score

REFRESH

Peg 15m / DEWS 30m

Preconditions & Failure Modes

MINIMUM DATA

PegScore requires ≥ 7 tracking days; DEWS requires ≥ 2 available signals (total weight ≥ 0.30) plus fresh core source tables

REQUIRED SOURCES

Peg events + tracking window inputs; DEWS consumes supply/liquidity/price plus optional flow/blacklist/yield signals

FAILURE BEHAVIOR

PegScore can be null; DEWS returns null when signal coverage is below threshold; stablecoins-cache failure aborts writes, while other source failures or stale DEX liquidity publish partial rows and mark the cron degraded

► Worked examples (verified against computePegScore and computeDEWS)

► Technical details: PegScore formula, DEWS signals, weights, and threat bands

Contagion Stress Test v1.0

The stress test simulates dependency failures to reveal systemic concentration risk across the stablecoin ecosystem.

See also: [Safety Scores](#)

SIMULATION ACTION

Force one coin to grade D

PROPAGATION CHANNEL

Dependency channel only

PRIMARY OUTPUT

Affected coins + supply at risk

Preconditions & Failure Modes

MINIMUM DATA

Target coin must have dependents and mapped dependency weights

REQUIRED SOURCES

Current report-card scores plus dependency map inputs

FAILURE BEHAVIOR

Only direct dependency-risk channel is recomputed (no peg/liquidity/confidence feedback loops)

► Worked example (verified against scoreDependencyRisk path used by stress test)

► **Technical details: simulation pipeline, scoreboard logic, and limitations**

Blacklist Tracker Methodology v3.993 [Version history →](#)

Version increments when tracked contracts, event parsing rules, cursor semantics, or amount-enrichment logic change.

The Blacklist Tracker monitors issuer intervention events across USDC, USDT, PAXG, XAUT, PYUSD, USD1, USDG, RLUSD, U, USDtb, A7A5, FDUSD, BRZ, AUDS, MNEE, EURI, USDQ, USDO, USDX, AID, TGBP, EURC, BUIDL, USDP, TUSD, NUSD, EURCV, USDA, USAT, AEUR, XUSD, XAUm, JPYC, FRXUSD, and FIDD contracts, including blacklist, unblacklist, block/unblock, account pause/unpause, and destroy/wipe actions across supported EVM and Tron networks.

Methodology revisions document changes to event coverage, cross-chain decoding behavior, cursor safety policies, event-time amount attribution rules, and the separate freeze-ledger snapshot used for the public summary and quarterly chart, including the reconciled `kyc.rip` / `stables.rip` bootstrap for ETH USDC, ETH USDT, and TRON USDT. Non-USD or commodity-denominated assets use coin-specific price-cache entries when Pharos reports USD frozen value.

Public frozen totals are last-known successful freeze-ledger snapshots rather than live balance guarantees. Provider refresh failures preserve the previous successful value and surface data-quality context. New snapshot identities are contract/config scoped, while older rows can use legacy symbol/chain/address fallback until remediation catches them up.

Blacklistability uses the report-card five-status model: Yes, Dilutable, Upstream, Possible, and No. Any reserve, custody, backing, parent-asset, or CEX/custody-rail exposure resolves as Upstream; Possible is reserved for curated direct token or vault controls that are not confirmed direct blacklist controls.

DATA SOURCES

Etherscan v2, chain RPC / dRPC log scans, and TronGrid

TRACKED EVENTS

Freeze, Unfreeze, Wipe (AddedBlackList, RemovedBlackList, DestroyedBlackFunds)

CHAINS

Ethereum, Tron, + supported EVM L2s

UPDATE FREQUENCY

Every 6 hours (`3 \*/6 \* \* \*`) + backlog reconciliation

Preconditions & Failure Modes

Minimum data	At least one indexed block range per chain
Required sources	Block explorer event logs with valid ABI decoding
Failure behavior	Preserves last-known snapshots; stale/provider-failed status shown

► **Worked example: blacklist event reconciliation**

► **Technical details: freeze ledger reconciliation**

Chain Health Score v1.2 [Version history →](#)

Version increments when factor weights, tier assignments, or sub-factor formulas change.

The Chain Health Score is a 0–100 composite that rates each blockchain's stablecoin ecosystem across five weighted factors. It answers: *how healthy, diverse, and resilient is the stablecoin mix on this chain?*

See also: [Liquidity Score](#)

SCORE RANGE

0–100 (null when safety-score coverage < 50%)

REFRESH CADENCE

15-minute stablecoins cache cadence; `/api/chains` freshness budget is 1800 seconds

DEPENDENCIES

DefiLlama supply, Pharos Safety Scores, peg rates

► **Formula & Weights**

► **Chain Resilience Tiers**

► **Health Bands**

► **Worked example: Ethereum vs PulseChain**

WHITE PAPER  **Download the full Pharos Methodology (PDF)**

Per-methodology PDFs are linked from each changelog page.